CLASS 11-CBSE MATHEMATICS

CHAPTER: STRAIGHT LINES-SET-1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. Write the equation of the straight line which passes through the point $(1,-2)$ and cuts off equal intercepts from axes.

2. If the lines $2x+3ay-1=0$ and $3x+4y+1=0$ are mutually perpendicular, then find the value of '*a*'.

3. Write the equation of the line perpendicular to the line $\frac{x}{a}-\frac{y}{b}=1$ and passing through the point at which it cuts *x*-axis.

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. Find the value of x for which the points $(x, -1)$, $(2, 1)$ and $(4, 5)$ are collinear.

5. Find the equation of the line which intersects the *x*-axis at a distance of 3 units to the left of origin with slope -2.

6. Find the distance between parallel lines

   (i) $15x + 8y - 34 = 0$ and $15x + 8y + 31 = 0$

   (ii) $l(x + y) + p = 0$ and $l(x + y) - r = 0$

7. Find the distance of the point $(-1, 1)$ from the line $12(x + 6) = 5(y - 2)$.

8. Find a point on the *x*-axis, which is equidistant from the points $(7, 6)$ and $(3, 4)$.

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. Without using distance formula, show that points (–2, –1), (4, 0), (3, 3) and (–3, 2) are vertices of a parallelogram.
10. The vertices of ΔPQR are P (2, 1), Q (–2, 3) and R (4, 5). Find equation of the median through the vertex R.
11. Show that the perpendiculars let fall from any point on the straight line $2x + 11y - 5 = 0$ upon the two straight lines $24x + 7y = 20$ and $4x - 3y - 2 = 0$ are equal to each other.

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. Find the distance of the point of intersection of the lines $2x + 3y = 21$ and $3x - 4y + 11 = 0$ from the line $8x + 6y + 5 = 0$.
13. If 3 lines whose equations are $y = m_1x + c_1$, $y = m_2x + c_2$ and $y = m_3x + c_3$ are concurrent, then show that

$$m_1(c_2 - c_3) + m_2(c_3 - c_1) + m_3(c_1 - c_2) = 0.$$

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